

Aim

To write a MATLAB program to find the convolution of two discrete time sequences.

Theory

In this program, we will take two sequences, namely $x[n]$ and $h[n]$ and compute their convolution:

We represent this as:

$$y[n] = x[n] \otimes h[n]$$

To do this we will perform four steps:

1. Flip: We will convert it as: $h[k] \rightarrow h[-k]$
2. Shifting: We will now shift the signal which we represent as follows: $h[-k] \rightarrow h[n-k]$
3. Multiplication: $x[n] \cdot h(n-k)$
4. Integration: $y[n] = \int_{-\infty}^{\infty} x[n] \cdot h[n-k]$

MATLAB Code

```
clear all
close all
clc

x = [1, 2, 3, 4];
h = [4, 3, 2, 1];

total_elements_in_y = length(x) + length(h) - 1;

x = [x zeros(1, total_elements_in_y-length(x))];
h = [h zeros(1, total_elements_in_y-length(h))];

y = zeros(1, total_elements_in_y);

for n = 1:total_elements_in_y
    for k = 1:n
        y(n) = y(n) + x(k) * h(n-k+1);
    end
end

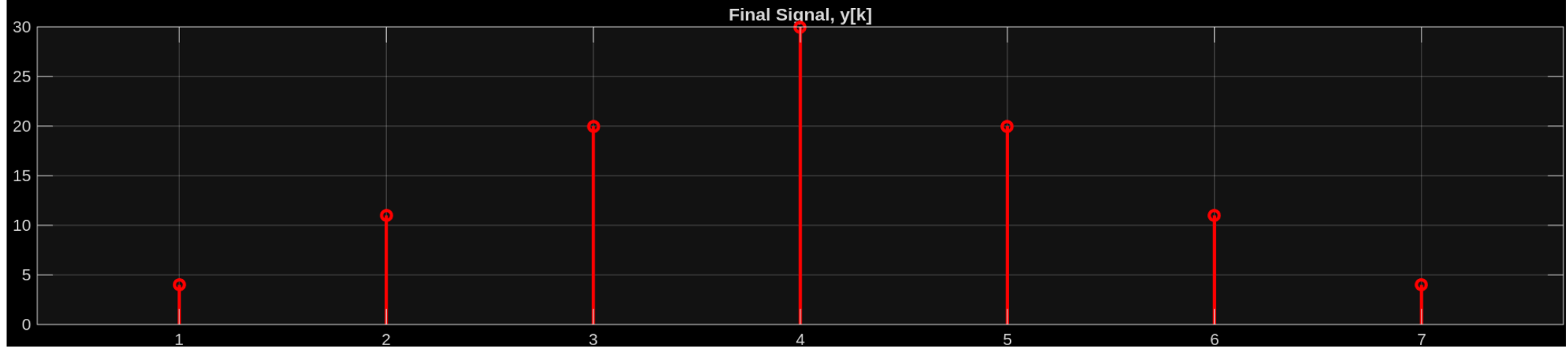
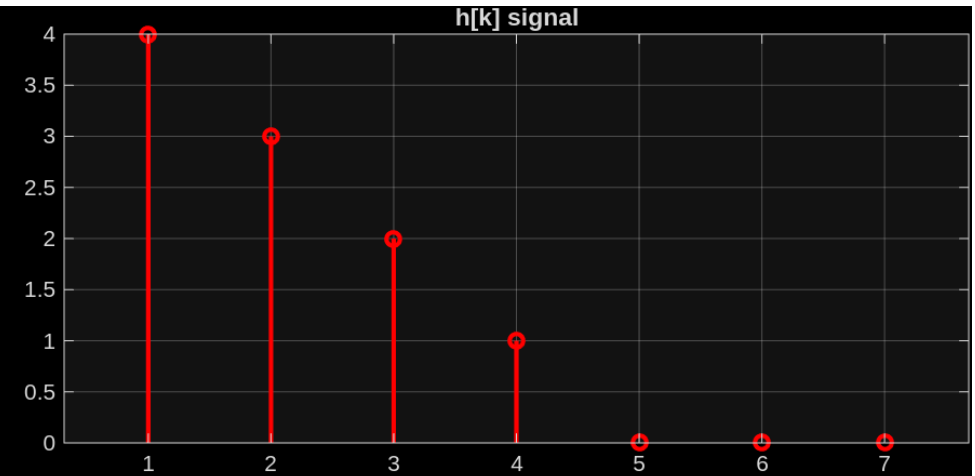
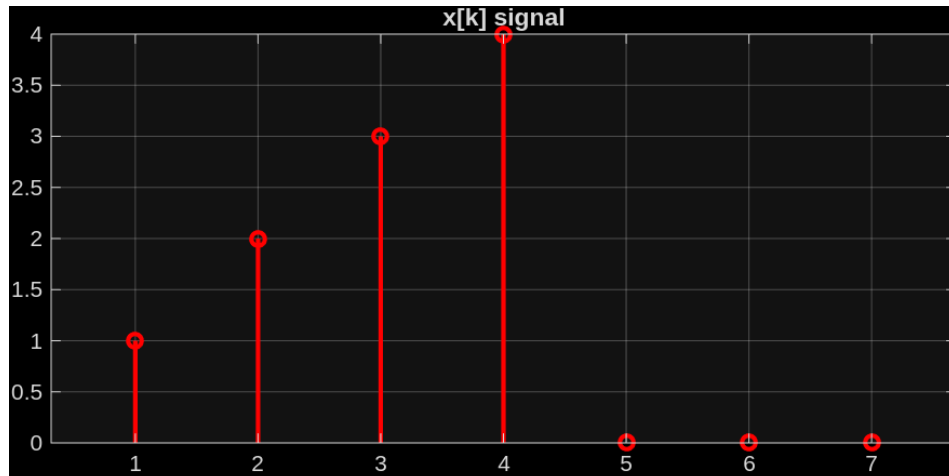
figure;

subplot(2,2,1)
stem(x, 'r',LineWidth=2)
grid on;
title('x[k] signal')

subplot(2,2,2)
stem(h, 'r',LineWidth=2)
grid on;
title('h[k] signal')

subplot(2, 1, 2)
stem(y, 'r',LineWidth=2)
title('Final Signal, y[k]')
grid on;
```

Results



Conclusion

In this experiment, we have successfully shown the computation of linear convolution between two discrete-time sequences. We found that if, for example,

$$x[n]=[1,2,3,4]$$

&

$$h[n]=[4,3,2,1]$$

We observe that the convolution of these signals yields:

$$y[n]=[4,11,20,30,20,11,4]$$